Alcoy (Alcoi), Alicante

The danger here is autumn flash floods and winter cold, not El Niño heat. The single most important action is a register of people who live alone or depend on powered medical equipment, with a named neighbour for each, built before the first orange warning.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Not resolved. Containing geography: Spain	No figure published for Spain or Europe	Sep 2026 – Feb 2027 (6-month)	Autumn flash flood (DANA), then winter cold, snow, and power loss

15/15 recommendations checked. 11 corrected, 2 unverified, 1 contraindicated, 1 confirmed as written. Exposure figure: Climate Impact Lab, *451,000 at Risk*, September 2026, pre-peer-review. It publishes **no figure for Spain or Europe**, so there is no national number to scale and none is given here. **The funding target fell by 58% in verification**, from €251k to €105k: one placement was contraindicated and another was cut to what the recipient has actually delivered before.

EXECUTIVE SUMMARY — AT A GLANCE



38.70°N 0.47°W, Spain · Natural Earth

- **The risk.** The danger is autumn flash floods, then winter cold and power loss, not El Niño heat: the 1986 Serpis floods peaked at 770 m³/s at Beniarrés, and January 2017 brought 159 l/m² and cut the A-7.
- **First action.** Serveis Socials, Creu Roja Alcoi, and the centres for older people build a register by 15 November of people alone or on home oxygen or dialysis, each with a named neighbour who knocks on an orange or red warning or a power cut.
- **The money.** €75k floor, €105k target, €145k ceiling; the envelope is set by what the town can spend before the peak, and the floor reaches 400 households at €150 each.
- **Do not.** Do not run a generator indoors or use candles to keep oxygen or other equipment going in a power cut: in the 28 April 2025 blackout, three people died of carbon monoxide in Taboadela.*

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* Suggested formulation: check with a health worker before sharing.

Read this first: why this is not an "El Niño winter" briefing. El Niño's link to eastern Spain is weak and changes with the season. The published work (Rodó, Baert and Comín, 1997; Shaman and Tziperman, 2011) associates it with *wetter autumns, September to December*. AEMET had published no winter 2026/27 forecast as of early October. Its October–December outlook is warmer than normal nationally, and ECMWF treats temperature as the most robust signal. The last El Niño (2023–24) brought Alcoy drought, not floods: 323.6 mm in 2023 and 326.2 mm in 2024, against 846.2 mm in 2022. The October 2024 Valencia DANA happened in ENSO-neutral conditions. **So the riskiest window is now through December, whatever the tropics do.** The actions below are keyed to that window and to winter, not to a tropical peak.

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the winter peak. Backing lenses are shown per action. An action several lenses reached independently is the stronger bet, but only where its evidence came from somewhere like here.

1. Build a register of people living alone and people on home oxygen or dialysis, and give each a named neighbour who knocks on an orange or red warning or a power cut

PARTIALLY CONFIRMED | EMERGENCY MANAGER · PUBLIC HEALTH · FRONTLINE · SKEPTIC · PHILANTHROPIST

ADAPTATION INVENTORY: NO SPANISH ENTRY; REGISTER PRECEDENTS IN JEREZ AND MAJADAHONDA

OWNER	BY WHEN	COST	PREVENTS
Serveis Socials (Dependència, teleassistència), with Creu Roja Alcoi and the centres for older people	15 November 2026	Staff time. Battery backup and CO alarms for the power-dependent: not costed	Deaths of people alone at street level in a flood, and of power-dependent patients in an outage

Alcoy has **3,604 people over 65 living alone, 29% of the age group** (INE data in the town's 2021 age-friendly diagnosis, which itself proposes extending teleassistance to everyone living alone). In the 29 October 2024 Valencia DANA, almost half the dead were over 70. In the later investigating court's (CID) count, about a third were found in homes and 14% in garages. In the 28 April 2025 blackout, three people in Taboadela (Ourense) died of carbon monoxide from a generator powering an oxygen machine, and a woman in Alzira died when her oxygen concentrator stopped. Spain has **no outage-priority register with the electricity distributor**. Ley 24/2013 art. 52.4 only bars

* Suggested formulation: check with a health worker before sharing.

disconnecting life-sustaining equipment, and a state electro-dependence registry exists only as a July 2026 draft decree. Jerez and Majadahonda built municipal registers after the blackout, which is the model to copy. Build the list from the teleassistance and home-help rolls and the health centre's home-oxygen supplier list, not from a door-to-door census, which cannot be finished in five weeks.

Transferability: the evidence that checking on people saves lives comes from heat (France after 2003, about 4,400 fewer deaths than predicted in 2006, with no single measure isolated), not from floods or cold. The death patterns it answers are Spanish and recent. The "Nos Motivamos" programme and the filaes' role could not be confirmed and are left out of the owner line.

2. Write the flood warnings now, in four languages, with "go up, not down" as the message, and a relay that does not wait for ES-Alert

PARTIALLY CONFIRMED | EMERGENCY MANAGER · SKEPTIC · FRONTLINE · PUBLIC HEALTH | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Protecció Civil d'Alcoi, with 112 CV; relay through neighbourhood associations, parishes, mosque, and Romanian church networks	**31 October 2026**	**Staff and volunteer time; leaflet printing**	**Deaths in garages, basements, and lifts, and on barranc crossings**

On 29 October 2024 the 112 CV centre had an ES-Alert drafted by about 18:37 and sent it at 20:11. The investigating judge found that by then "practically all" deaths had already occurred, and the message told people to avoid travel when most victims were at home. Alcoy's municipal flood plan (PAM d'Inundacions) was approved on 14 July 2023. Get its text and add pre-approved warning texts and release rules tied to **AEMET orange and red rain warnings and to observers at the Molinar, Barxell, and Riquer channels**. Then issue a one-page leaflet in Valencian, Castilian, Arabic, and Romanian: never go down to garages, basements, lifts, or underpasses to move a car; go upstairs; never cross a barranc. Alcoy has 7,846 foreign residents, the largest groups from Morocco (2,099) and Romania (1,372). Each WhatsApp phone tree needs one person whose job is to ring landlines.

* Suggested formulation: check with a health worker before sharing.

Correction: the CHJ publishes **no river-flow gauge in Alcoy**. The nearest Serpis gauges are downstream (Beniarrés reservoir and below), too far to warn the town, so gauge thresholds are not feasible by 31 October. Rain warnings and human observers are what is available. The foreign-resident figures come from local press and were not checked against INE.

3. Enrol every eligible household in the bono social before 31 December, and route the severely vulnerable through Servicios Sociales

PARTIALLY CONFIRMED | ECONOMIST · PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Ajuntament (Servicios Sociales), with Cáritas and Creu Roja as referral points	Enrolled by 31 December 2026	Staff time. The payers are national schemes	Cold homes and supply cut-offs

The trigger here is a date, not a rainfall reading. Households on the bono social eléctrico on 31 December receive that year's Bono Social Térmico automatically (RDL 15/2018). It is paid by IVACE+i from Ministry for Ecological Transition (MITECO) funds. The electricity discount is 42.5% for vulnerable and 57.5% for severely vulnerable households, under RDL 7/2026, **only until 31 December 2026**, and it reverts to 35% / 50% in 2027 unless extended. The ban on cutting off water, electricity, and gas for vulnerable consumers is also in force until 31 December 2026. It lapsed between about 28 January and 21 March 2026, after Congress rejected RDL 16/2025. It protects households classed as vulnerable under RD 897/2017, which in practice means bono social holders. A severely vulnerable household whose bill Servicios Sociales part-pays is protected permanently.

* Suggested formulation: check with a health worker before sharing.

Corrections: the Térmico averages about **€139 per beneficiary in the Comunitat Valenciana** (our division: €27M ÷ 193,629 for the 2025 round; 2024 gives the same), not the €400 one source reported. It arrives months after the winter: the 2025 round was paid in June 2026, and 23,178 payments failed on bad bank details. Registering with Servicios Sociales does not by itself trigger the cut-off ban. Collect correct bank details at enrolment.

4. Close the flu and COVID vaccination gap in over-65s living at home, and push RSV uptake in care homes

PARTIALLY CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Departamento de Salud de Alcoy (primary care), with the register from action 1	**Mid-December 2026**	**Within the Conselleria's campaign; outreach is staff time**	**Hospitalisation and death in the flu season, which drives winter excess deaths in Spain**

The Conselleria's joint flu, COVID, and RSV campaign is already running, so this is not a new programme. The action is closing the gap in people who are not reached. Flu vaccine effectiveness against hospitalisation in over-65s in the Valencia region was **46.5% (95% CI 16.5–65.8) in 2018/19** (Mira-Iglesias et al., IJERPH 2021). In a weak season (2016/17, people over 60) it was 19%, with a confidence interval crossing zero (Eurosurveillance). Coverage in 2025/26 was 83.8% in care homes. In over-65s it was about 50.5% mid-campaign, an industry estimate (Sanofi Gripómetro) and not final, against a 75% target. RSV vaccine is offered **only to care-home residents and immunocompromised people**, not to everyone over 80.

Transferability: strong, as the evidence is from this region. The 2026/27 start dates could not be confirmed from an official Conselleria release.

5. A cold sweep of over-80s living alone: certified electric oil-filled radiators, CO alarms, and bono social enrolment

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE · EMERGENCY MANAGER | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Serveis Socials and primary care, using the action 1 register	**30 November 2026**	**Not costed; see the funding table for the voucher line**	**Cold-related deaths and CO poisoning**

* Suggested formulation: check with a health worker before sharing.

In Spain, the spread of home heating explains about **38% of the decline in cold-related deaths between 1989–93 and 2009–13, and about 51% for extreme cold** (Achebak et al., Environment International 2023, 48 provinces plus the Balearics, 1980–2018). That is province-level ecological evidence. It does not show that heating "halves" deaths. A Barcelona façade-retrofit study (Peralta et al., Int J Epidemiol 2017) found a borderline protective effect in women (RR 0.46, 0.21–1.01) and none in men, and retrofits cannot be built before this winter anyway.

Safety: hand out only certified electric radiators, and pair every visit with a CO alarm. Do not hand out butane heaters or anything that encourages braseros. Check the wiring in older flats. This safety guidance is the verifier's own judgement, not from a cited source. Centro-Viaducto's share of single-person households (over 40%) comes from a 2012 plan and is dated.*

6. Clear debris from bridge and culvert openings on the Barxell, Molinar, and barrancs, and file the CHJ authorisation this week

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Municipal works brigade, with Confederación Hidrográfica del Júcar (CHJ) authorisation	15 November 2026	Existing crews	Debris dams at bridges and overflow into streets

The CHJ's own Barxell channelling project cites the river's "frequent overflows" and an undersized bridge. The Serpis peaked on 1 October 1986 at 770 m³/s at Beniarrés. Work in the channel needs CHJ authorisation, so apply now. Start a first-pass audit of storage, parking, and ground-floor uses that sit in the channels; a full audit will not be finished in five weeks.

* Suggested formulation: check with a health worker before sharing.

Correction: the original basis cited the 30 September 1997 storm and 1987 channelling. The 1997 event hit Alicante city, and the November 1987 floods hit the coast (817 mm at Oliva). Neither is documented in Alcoy, so both are removed from the basis. No source was found for industry leaving Alcoy's riverbanks after 1986.

7. Add Alcoy-specific pieces to the existing winter road plans: snowploughs staged on the A-7 Alcoi–Ibi stretch, agreed closure criteria, and resupply priority for pharmacies and dialysis

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Ajuntament asks the Ministerio de Transportes (Demarcación de Carreteras), the Diputació, the DGT, and FEDAC	30 November 2026	Within existing winter road contracts	The town being cut off, as in 2017 and 2020

The ministry already runs a November–April winter road campaign (Vialidad Invernal) on the A-7, and the Diputació runs its own on provincial roads, so the action adds to those plans rather than building a new one. On 16–23 January 2017 AEMET recorded 159.0 l/m² at Alcoi, the A-7 was cut between km 450 and 464, and nine CV roads in the Alcoi sector closed. During Storm Gloria in January 2020 the town was cut off from the motorway, and the business federation FEDAC said a snowplough staged on the Alcoi–Ibi stretch would have kept it open.

* Suggested formulation: check with a health worker before sharing.

Corrections: the 2017 storm dates are 16–23 January, not 17–23. The 2017 Alcoi–Xàtiva rail cut was announced but is not confirmed to have happened. The 1926 snowfall claim could not be sourced and is dropped.

8. Pre-approve a fast-track assessment for emergency social aid that opens when AEMET issues an orange cold or rain warning

UNVERIFIED AS PROPOSED | ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Ajuntament d'Alcoi, Servicios Sociales	30 November 2026	Budget line not found	Delay between a warning and money reaching households

The Economist proposed automatic release on a warning. Emergency social aid in the Comunitat is granted only after a social worker assesses each household, so **automatic payment is unlikely to be lawful**, and no trigger-based scheme was found anywhere in Spain. The workable version is an assessment step activated by the warning. Alcoy's annual household-supplies aid shows the town can run renewals at scale: in 2021 it gave €103,815 to 1,306 recipients, 1,067 of them renewed automatically, and €119,192 in fee rebates to 1,269. The aid is small (€75–120 a year) and is not emergency aid.

The figures are from 2021, so ask the Ajuntament for 2025 numbers.

9. A "check your home insurance" campaign for ground-floor homes along the barrancs, before the autumn storms

CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Ajuntament, with neighbourhood associations	31 October 2026	Leaflets; the premiums are paid by each household	Uninsured flood loss, which falls hardest on poorer households

Spain's state-backed Consorcio de Compensación de Seguros pays extraordinary flood damage **only where a private policy exists**. Nationally 80.8% of homes are insured. Valencia province was at 78.4% before the 2024 DANA and 84.5% at the end of 2025. The Consorcio had paid about €3.6bn for that DANA by September 2025, passing €4bn by November. No figure exists for Alicante province or

* Suggested formulation: check with a health worker before sharing.

for Alcoy.

Protects assets, not lives. Ranked here because it is cheap, and an uninsured household that loses its ground floor is the one that cannot afford heating afterwards.

10. For 2027: fuel and erosion work in the Font Roja and Mariola this winter, and drought triggers written into the new water contract

PARTIALLY CONFIRMED · HISTORIAN · FRONTLINE · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Generalitat forestry, Ajuntament; the water operator under the new concession	Before spring 2027; the tender is open now	Not costed	Spring fire and water shortage

The July 2025 Ibi–Alcoi fire burned 185 ha, 147 of them in the Font Roja park, and a University of Alicante and CEAM study says the burned pine needs erosion control. The water service is run by FCC Aqualia on an expired concession, and a new 15-year, €62.2M contract is out to tender now, with five bids. That is the moment to write drought extraction triggers for the Molinar aquifer into it. Its record low was 37.89 m in February 2003. No confirmed winter schedule for fuel works in the Font Roja or Mariola was found.

There is a published autumn-wet signal, but no confirmed dry-spring signal behind this action. It rests on the 2023–24 drought record, not on the ENSO forecast.

11. For summer 2027: fund the heat plan the council agreed in 2023, and keep the Àgora refuge open in the afternoon

PARTIALLY CONFIRMED · PUBLIC HEALTH · PHILANTHROPIST · ADAPTATION INVENTORY: PRECEDENT (ITALY HEAT-HEALTH WARNING SYSTEM)

OWNER	BY WHEN	COST	PREVENTS
Ajuntament d'Alcoi	Budgeted by spring 2027	Not costed; no budget exists	About one heat death a year at Alcoy's scale

* Suggested formulation: check with a health worker before sharing.

Heat prevention plans are associated with a **25.2% fall in heat-attributable deaths across Europe (95% CI 19.8–31.9) and 15.3% in Spain (7.2–23.0)**, or 1.8 avoided deaths per 100,000 a year (Urban et al., Environmental Research Letters, December 2025, 14 countries). For Alcoy that is about one death a year. The Adaptation Inventory's nearest precedent is Italy, where the 2004 heat-health warning programme cut heat deaths by more than 57% on days at 30 °C or above (Pavanello and Valenti, 2025, working paper, not peer-reviewed). Alcoy agreed a heat plan in July 2023, but it is still not drafted and has no budget. A June 2026 heat-alert protocol names the Centre Cervantes Jove (8:00–20:00) and the Àgora (8:00–14:00) as climate refuges, and the Àgora closes before the hottest hours.

This falls after the winter peak, when the event decays. The claim that the plan "began in August 2025" could not be confirmed.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Serveis Socials	Protecció Civil & Policia Local	Ajuntament (works, budget, legal)	Generalitat (112 CV, Conselleria, forestry)	Departamento de Salud de Alcoy	Cáritas	Creu Roja Alcoi	Neighbourhood & faith networks	CHJ, Ministerio, Diputació, DGT	Residents
1. Register and named neighbour	**A/R**	**C**			**R**		**R**	**R**		**I**
2. Flood warnings and relay	**C**	**A/R**		**C**				**R**		**I**
3. Bono social by 31 December	**A/R**					**R**	**R**			**R**
4. Vaccination gap	**C**			**C**	**A/R**					**I**
5. Cold sweep: radiators, CO alarms	**A/R**				**R**	**C**				**I**
6. Culverts and bridge openings		**C**	**A/R**						**C**	
7. Snow access on the A-7		**C**	**A**		**C**				**R**	
8. Fast-track emergency aid	**A/R**		**C**							
9. Check your home insurance			**R**					**R**		**A/R**
10. Fuel works, erosion, drought triggers			**A/R**	**R**						
11. Fund the 2027 heat plan	**R**		**A/R**		**C**					**I**

What the chart shows. **Serveis Socials is accountable for five of the eleven rows**, and four of those depend on the register in row 1, so that one list is the bottleneck for the whole winter. **The two flood rows sit with Protecció Civil and the works brigade**, and both need someone outside Alcoy to act: 112 CV to send, the CHJ to authorise. **Residents are accountable for one decision only they can make**: whether their ground floor is insured before the storms.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per euro. The envelope is bounded by what this town can spend before the peak, not by what it needs.

FLOOR

€75k: 400 households × €150 + €15k

TARGET

€105k: 600 households × €150 + €15k

CEILING

€145k: target + €40k, only if Creu Roja confirms a non-duplicating programme

LARGEST UNCERTAINTY

Whether Cáritas can scale beyond the ~490 cylinders of its largest campaign

* Suggested formulation: check with a health worker before sharing.

The lens proposed a €251k target. Verification cut it by 58%, which is more than a third. The €196k heating line assumed 1,306 households, but that figure counts the municipal aid grants of 2021 (paid from an active budget line, which vouchers would duplicate), and it was about 20 times Cáritas's largest recorded campaign. It is cut to 600 referred households, which matches the 600-plus families a month Cáritas's shared shop served in 2014. "Boilers on prescription" (€80k) is contraindicated, as shown below. The ceiling's original cap of twice the town's €223k yearly aid throughput counted €119k of fee rebates that are not cash. Cash aid of about €104k × 2 gives a cap of about €208k, and €145k sits under it. Expected deaths averted are in the low single digits, so no honest cost per death can be given.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Cáritas Interparroquial d'Alcoi	Heating vouchers for referred households, paired with CO alarms (actions 3 and 5)	€90k (600 × €150)	Existing parish voucher-to-distributor channel; a top-up, not paid to households already receiving the municipal supplies aid	Bridges the gap until the ~€139 Bono Social Térmico arrives, months after the winter. Cáritas campaigns bought 490 cylinders in 2017 and 260 in 2019. No published cost per death	PARTIALLY CONFIRMED
Serveis Socials (Ajuntament)	Register of people living alone and power-dependent residents (action 1)	€15k	Direct grant	The lens's own bottom-up estimate, unsourced. Heat plans overall avert 1.8 deaths per 100k a year (Urban et al., 2025), but that is for whole plans, not registers	PARTIALLY CONFIRMED
Creu Roja Alcoi	Cold-season phone calls and home visits for over-65s living alone	€40k (ceiling only)	Grant, conditional	One coordinator for four months, plus nurse visits and volunteer expenses. No spending history found, and Cruz Roja teleassistance already includes scheduled calls elsewhere, so overlap is likely	UNVERIFIED

The funder's one thing: the money here is a timing problem. The state already pays a heating subsidy, but it lands in June. The cheapest life saved is the one where that committed money arrives before the cold, through a channel (Cáritas's parish vouchers) that already works, at the scale it has worked at.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for these hazards and would cost lives here, or would waste the window.

Running a generator indoors, or using candles, to keep oxygen or other equipment going in a power cut

CONTRAINDICATED

In the 28 April 2025 Iberian blackout, three people in Taboadela (Ourense) died of carbon monoxide from a borrowed generator running indoors to power an oxygen machine. A candle fire killed a woman in Madrid. The 10 August 2026 storm brought gusts of 115 km/h to Alcoy, so outages here are not hypothetical.

Instead: put power-dependent residents on the action 1 register, supply battery backup sized to their equipment, fit CO alarms, and plan a move to a powered site before the battery runs out.*

Waiting for the ES-Alert, or going down to the garage to move the car

CONTRAINDICATED

On 29 October 2024 the alert came at 20:11, after practically all the deaths. It told people to avoid travel, when many died at home, in garages, ground floors, and lifts, often trying to save their cars. About 14% of those found were in garages (CID count).

Instead: act on the AEMET orange or red warning and on what you see in the channels. Go up, not down. A car is not worth a life.*

Funding "boilers on prescription" home repairs this winter

CONTRAINDICATED

The cost basis was two UK figures from different schemes. Bray et al. (2017) studied a combi boiler plus double glazing in 228 North East England social-housing homes, and gave no cost per QALY. The £17,300/QALY is Worcestershire County Council's own estimate for a different scheme. Most homes in Alcoy have no wet central heating and run on butane or electric heaters. Referral, procurement, and

* Suggested formulation: check with a health worker before sharing.

installation would not be finished before the peak.

Instead: certified electric radiators and CO alarms through the action 5 sweep, and vouchers through Cáritas.

For summer 2027: handing fans to over-65s as the heat measure

CONTRAINDICATED

A 2024 critical review (Global Heat Health Information Network) finds fans give no meaningful core cooling for people aged 65 and over above about 35 °C.*

Instead: access to a cooled room, and keep the Àgora refuge open through the afternoon.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Late Sep – 1 Oct 1986	Serpis floods; peak 770 m³/s at Beniarrés (El Niño onset year)	No Alcoy toll found	Not documented locally
Feb 2003	Molinar aquifer record low, 37.89 m	None	Passed without a supply crisis
16–23 Jan 2017	Rain, snow, and wind: 159.0 l/m² at Alcoi; A-7 cut km 450–464; nine CV roads closed	No deaths reported	Nothing documented
19–22 Jan 2020	Storm Gloria: 287.2 l/m² at Muro d'Alcoi, 471.2 at Beniarrés; town cut off from the A-7	An elderly woman (reported 70–75) died when a building on Calle San Agustín collapsed in the rain	FEDAC criticised the road closure; no plan change found

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2023–2024	Drought during the last El Niño: 323.6 mm and 326.2 mm, against 846.2 mm in 2022	None recorded	Nothing documented
14 Jul 2023	Council approves flood, seismic, and forest-fire plans (PAM)	—	Heat plan agreed the same month, still unbudgeted
29 Oct 2024	Valencia DANA (not Alcoy; the benchmark failure). ENSO-neutral	About 229 dead in Valencia province	Court case on the late ES-Alert; Unespa insurance penetration in Valencia rose from 78.4% to 84.5%
Jul 2025	Ibi–Alcoi fire: 185 ha, 147 in the Font Roja park	No deaths; scattered houses evacuated	Erosion control recommended
10 Aug 2026	Storm: gusts of 115 km/h (AEMET Alcoi), trees down	No deaths reported	—

The recurrence nobody has noticed: in Alcoy, the deaths and the isolation come from *winter* storms, not summer heat. Gloria killed a woman when an old building gave way in prolonged rain, and both 2017 and 2020 cut the town off on the A-7. The Historian saw that the floods of 1986–87 and 1997 fell in El Niño years. Verification found that overstated: only 1986 is a documented Serpis flood, the 1987 and 1997 storms hit the coast and Alicante city, and three events chosen after the fact do not establish a link. Treat ENSO as background, not as a trigger.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **"Frame this as El Niño preparedness"** versus **"Frame it around the DANA season and winter."** The Historian read an El Niño link into Alcoy's flood record. The Skeptic warned that a campaign keyed to a tropical winter peak arrives after the riskiest months, and would read as a false alarm if the winter turns dry, spending credibility the next real DANA warning needs. Verification sided with the Skeptic. This briefing is framed accordingly.
- **Butane vouchers through Cáritas** versus **the CO risk of combustion heating for frail people.** The Philanthropist's best channel delivers butane, while the Skeptic's documented killers include CO. Resolved here by pairing every voucher with a CO alarm, but nobody has evidence

* Suggested formulation: check with a health worker before sharing.

on the net effect.

- **Money released automatically on a warning** (Economist) versus **the legal requirement for a social worker's assessment** (verification). It would be settled by the Ajuntament's legal services on a fast-track design.
- **Pay Creu Roja for winter calls** (Philanthropist) versus **teleassistance probably already makes them** (verification). It would be settled by asking Creu Roja Alcoi and Dependència what calls they already schedule.

WHAT NOBODY COVERED

BLIND SPOT

The one recorded weather death in Alcoy in the last decade was a **building collapse in rain**, and no lens built an action around the condition of the old textile-era housing stock. The Economist suggested inspections but found no cost or owner, so it was not carried forward. Nobody looked at **care homes** either: how many there are in Alcoy, and whether they have backup power and staff cover for a snow-isolated week. Either could change the ranking: if the old centre has buildings already flagged as unsafe, inspecting them before the winter rains could outrank actions 6–9.

HOW TO READ THIS

METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people, and none is published for Spain. For scale only, the Public Health lens noted that Spain's 2022 summer heat-death rate (Ballester et al., Nature Medicine 2023, about 237 per million) applied to Alcoy's population gives roughly 14 deaths in a bad summer. That is our arithmetic, not an Impact Lab finding and not an El Niño projection. The El Niño projection covers *heat only*; flood, cold, snow, fire, and drought are carried here by the historical record rather than by a published number.

* Suggested formulation: check with a health worker before sharing.

- **Scope.** The location was taken as Alcoy (Alcoi), the city of about 62,000 in the comarca de l'Alcoià, Alicante province, Comunitat Valenciana. The requester typed "Alcoy", and confirmed by email that they meant the Spanish city, not Alcoy in Cebu, Philippines. Climate Impact Lab impact region: not resolved. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); neighbourhood and street detail is used for hazard exposure and action design, not for death counts. At about 62,000 people, the town is itself the right unit for most actions.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Heat is not the dominant hazard here, so the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted by reading the dataset embedded in the inventory page. It has no entries for Spain. The nearest Mediterranean precedent is Italy's heat-health warning system (cited in action 11). Nothing in it covers flood warning, cold, snow access, or power-dependent patients.
- **Health wording.** Passages where a wording error could hurt someone (heaters and carbon monoxide, medical equipment in a power cut, flood warnings, and fans for older people) are marked "* Suggested formulation: check with a health worker before sharing." They were

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checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so. The ~€139 Bono Social Térmico average is a division of published regional totals and is labelled as such. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in Alcoy was interviewed. The impact region was not resolved because the Impact Map is client-side. The text of the 2023 municipal flood plan was not obtained. No Alcoy-specific data was found on heating or air-conditioning prevalence, energy poverty, care homes, or how foreign residents are spread across neighbourhoods. Several local figures rest on press reports (El Periòdic, COPE Alcoy, Alicante Plaza) rather than official releases. The judge's finding on the ES-Alert is taken from media quoting the court order, not the order itself.

References · verified unless marked

ENSO and season: Shaman & Tziperman 2011 · Rodó, Baert & Comín 1997, Climate Dynamics (abstract only) · El Español on the AEMET Oct–Dec outlook

Population and ageing: Alcoi age-friendly diagnosis 2021 · El Nostre Ciutat, foreign residents (not checked against INE)

DANA 2024: elDiario.es on the court order · Valencia Plaza, where victims were found · Congreso DANA report (death toll; not re-checked)

Power dependence: Defensor del Pueblo · Sanidad draft decree, Jul 2026 · El Español, blackout deaths · Jerez register

Flood plan and gauges: PAM approval 2023 · CHJ SAIH gauges · CHJ Barxell project · Serpis 1986 peak (secondary)

Finance: RDL 15/2018 · Pérez-Llorca on RDL 7/2026 · Qué!, Bono Térmico 2025 round (its €400 figure is wrong) · Bono Térmico 2024 · Alcoi supplies aid 2021 · Unespa insurance penetration · Moncloa, Consorcio payouts

Health: Mira-Iglesias et al. 2021 · Eurosurveillance 2016/17 · RSV in care homes · Achebak et al. 2023 · Urban et al. 2025 · Pavanello & Valenti 2025 (working paper) · Fans review · Alcoi heat protocol 2026

Funding: Cáritas 2017 campaign · Cáritas 2019 campaign · Bray et al. 2017 · Worcestershire

Record: AEMET Jan 2017 · FEDAC on Gloria · CHJ Gloria report · Gloria deaths · COPE Alcoy rainfall · Molinar aquifer · 2025 fire · Water contract tender · Vilaweb, Aug 2026 storm

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 10 October 2026 for Alcoy (Alcoi), Alicante, Spain. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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